



AALBORG UNIVERSITY
DENMARK

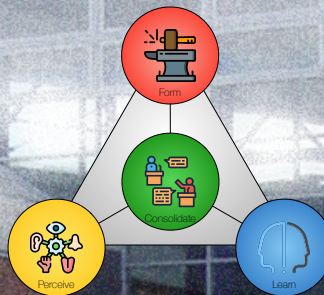
10 TACTICS, MANIFESTATIONS, CAPABILITIES & GUEST LECTURE

ESSENCE – CHAPTERS 15, 16 & 17 (SLIDES ONLY)

GUEST LECTURE: KLAUS BUNDVIG, NETCOMPANY

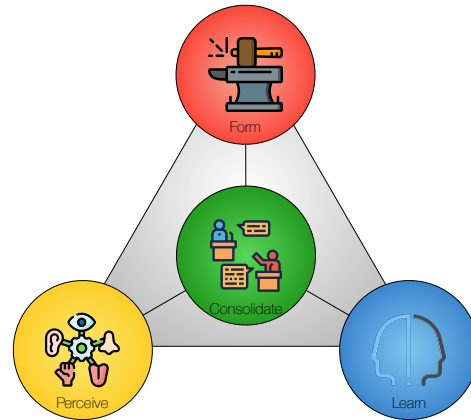
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Agenda

- Exercise 9:
 - Evolvability & General Innovation Theory.
- Lecture 10:
 - Chapter 15 in Essence: Tactics (**slides only**).
 - Chapter 16 in Essence: Manifestations (**slides only**).
 - Chapter 15 in Essence: Capabilities (**slides only**).
 - Guest lecture: **Klaus Bundvig, Netcompany.**

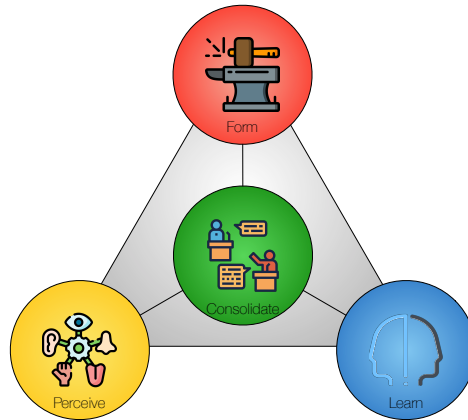


Follow-up: Exercise 9

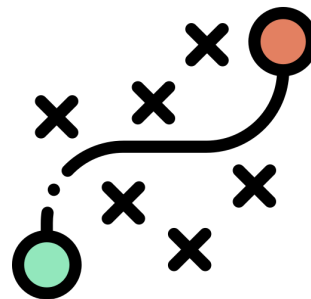
Exercise 9 (Evolvability & General Innovation Theory)

This exercise concerns the **evolvability** and **innovation characteristics** of your project.

- Describe and illustrate **diffusibility** in relation to your project.
- Describe and illustrate **adoptability** in relation to your project.
- Evaluate your project using one or more of the **evaluation types** presented on the slides.
- Describe your project using the terms **invention**, **innovation**, **exploitation**, and **diffusion**.
- Describe the **novelty**, **radicality**, and **timing** of your project.



Tactics



Do the difficult things while they are easy and do the great things while they are small.

A journey of a thousand miles must begin with a single step

Lao Tzu (6th century BC–5th century BC)

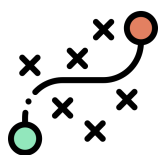
Three Levels of Abstraction



Rationale – the logical basis that makes actions meaningful and helps reason about strategy and tactics for solving the overall problem.



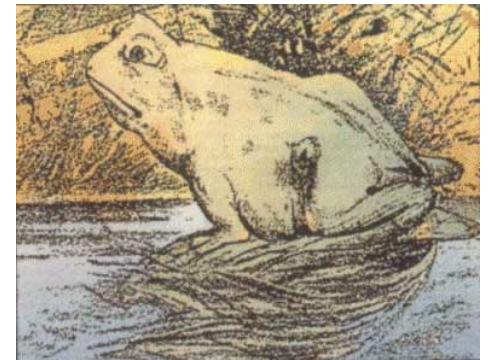
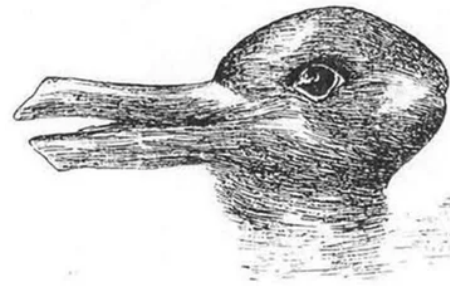
Strategy – the **overall plan** for addressing the problem, including its **scope** and **context**, the essential parts needed to build the solution, and any limitations or constraints that might influence its feasibility. At the strategy level, we also evaluate the potential for future **growth**.



Tactics – actions planned in accordance with the strategy to achieve specific ends.

Observation is Theory-laden

- Norwood Russell Hanson (1924 – 1967) was an American philosopher of science.
- He argued that observation is theory-laden. **Observation language** and **theory language** are deeply connected.
- What we **perceive** is not what our **senses** receive, but is **filtered** through our existing preconceptions.
- We cannot separate **sensing** and **interpreting**.
- For that reason, we see the **abstract** and the **concrete** as **complementary** in inquiry and resolution.



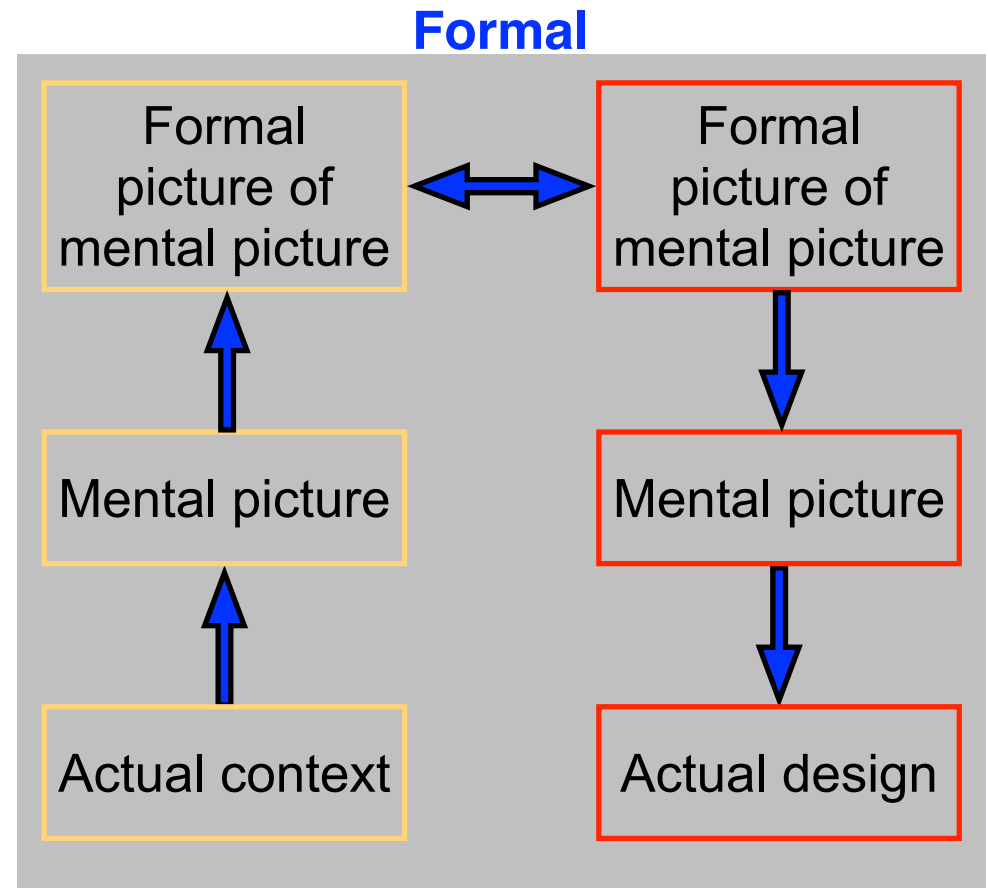
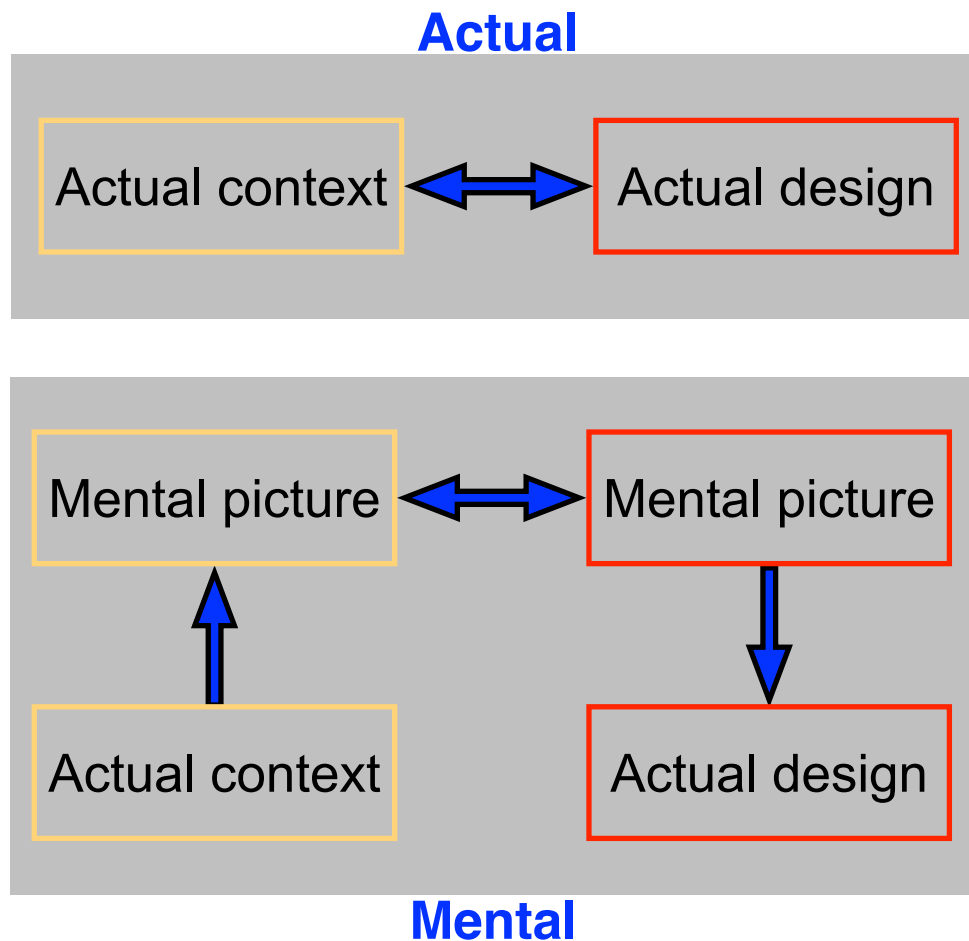
Principles of problem formulation and solutions

- The philosophical foundation is Deweyan pragmatism, drawing on terms such as **transaction**, **inquiry**, and **situation**.
- The work sequence can begin with **identifying the problem** and then finding a solution, or **start with a solution** and determine appropriate problems.
- The purpose of the **problem formulation principles** is to strike a **balance** between the **concrete** and the **abstract**.
- Similarly, the purpose of the **solution principles** is to strike a balance between the **concrete** and the **abstract**.
- A **concrete** formulation of a **problem** helps develop **solutions** that can be tested and confirmed.
- An **abstract** formulation of a **problem** helps develop **solutions** that are general and can be adjusted to address **problems that are similar**, in whole or in part.
- At the **concrete** level, the problem manifests as a set of **manifestations**.
- Similarly, at the **concrete** level, the solution is detailed in terms of **capabilities**.

Problem and solution formulation are linked

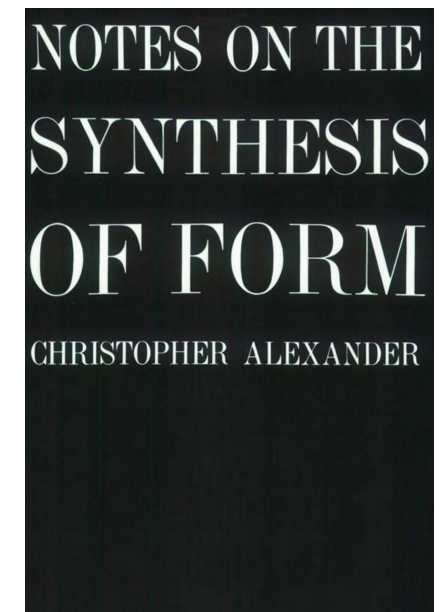
- Moving between the concrete and the abstract.
- Moving between the problem side and the solution side.
- Moving between the current situation and future options.

Actual, Mental, and Formal Views on Situations and Designs



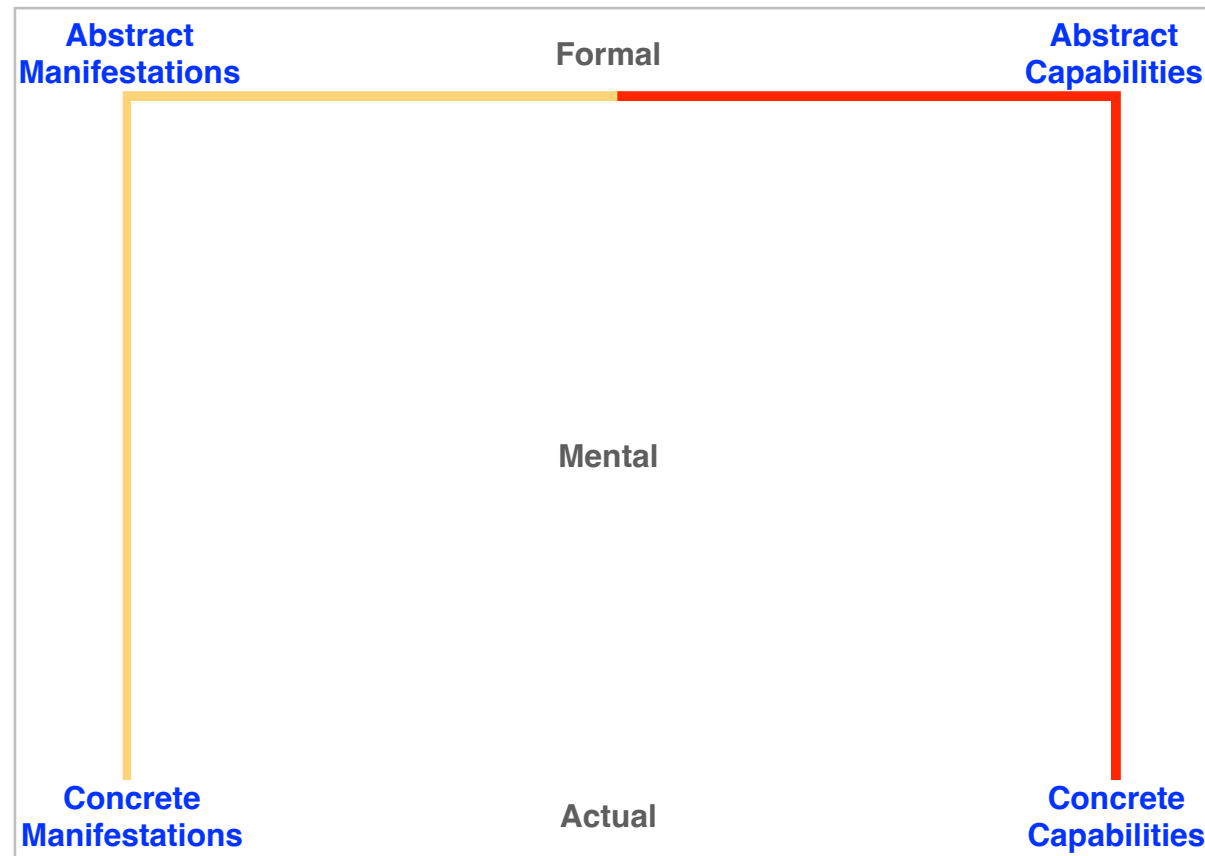
The Unselfconscious, the Selfconscious, and the Qualified

- The **unselfconscious** “designer” works only at the **level of the actual**. The problem-solver simply reacts to misfits by changing them.
- The **selfconscious** designer works also at the **mental level**. Design is shaped intuitively by a **conceptual** interaction between a mental picture of the context and a repertoire of ideas.
- The **qualified** designer also works on the **formal level**. Design is shaped by selecting **patterns** based on their pros and cons.

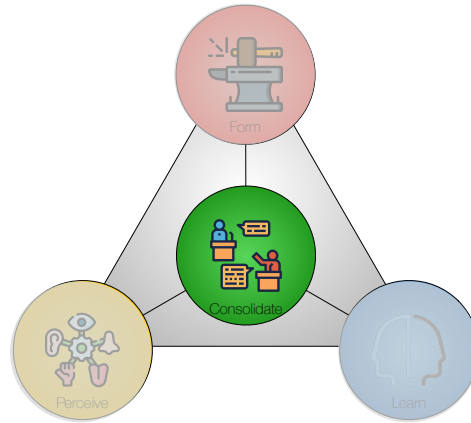


First out 1964

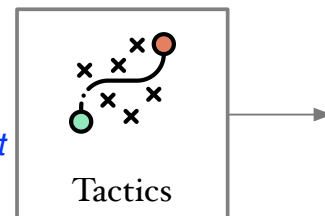
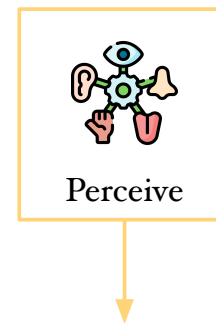
The **abstract** and the **concrete** are complementary



Inquiry is a movement between the **situation**, the **problem**, and the **solution**.
Push or Pull?—Inquiry may begin on either the problem or the solution side.



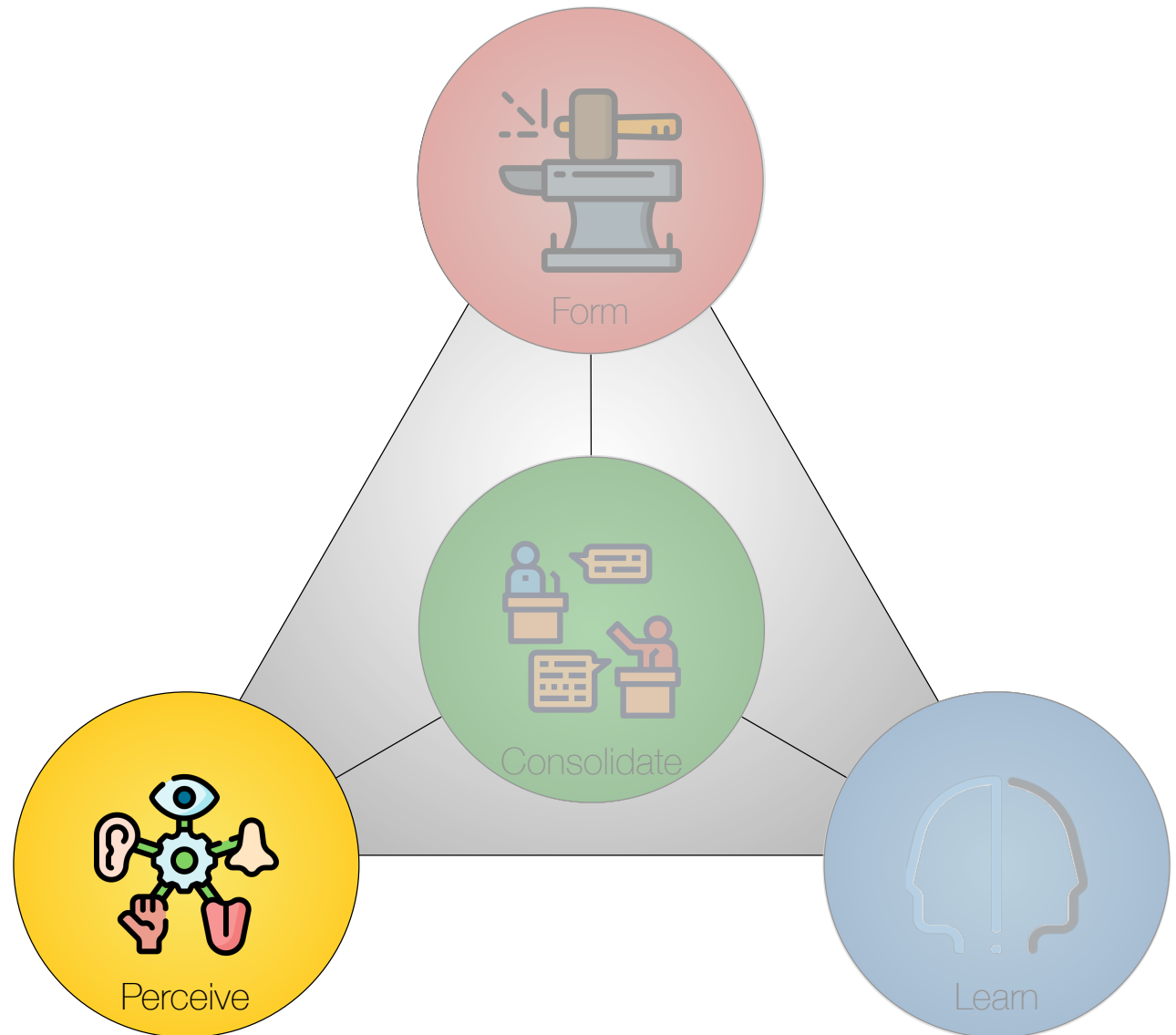
Manifestations



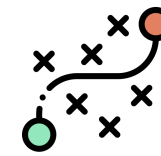
Solving a problem simply means representing it so as to make the solution transparent.
Herbert Simon (1916–2001)

Four Core Activities

- *Perceive* – understanding the problem and its context.
- *Form* – designing the constructive parts of a solution.
- *Consolidate* – integrating the design contributions into a clear whole that addresses the situation.
- *Learn* – appraising the design.



Perception Terminology



Problem



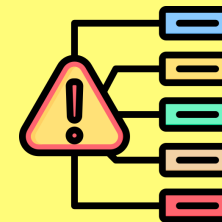
We see a problem as a situation to be handled, where problems appear as manifestations.

Outer Environment



The problem is found in the outer environment. The design is adapted to this domain.

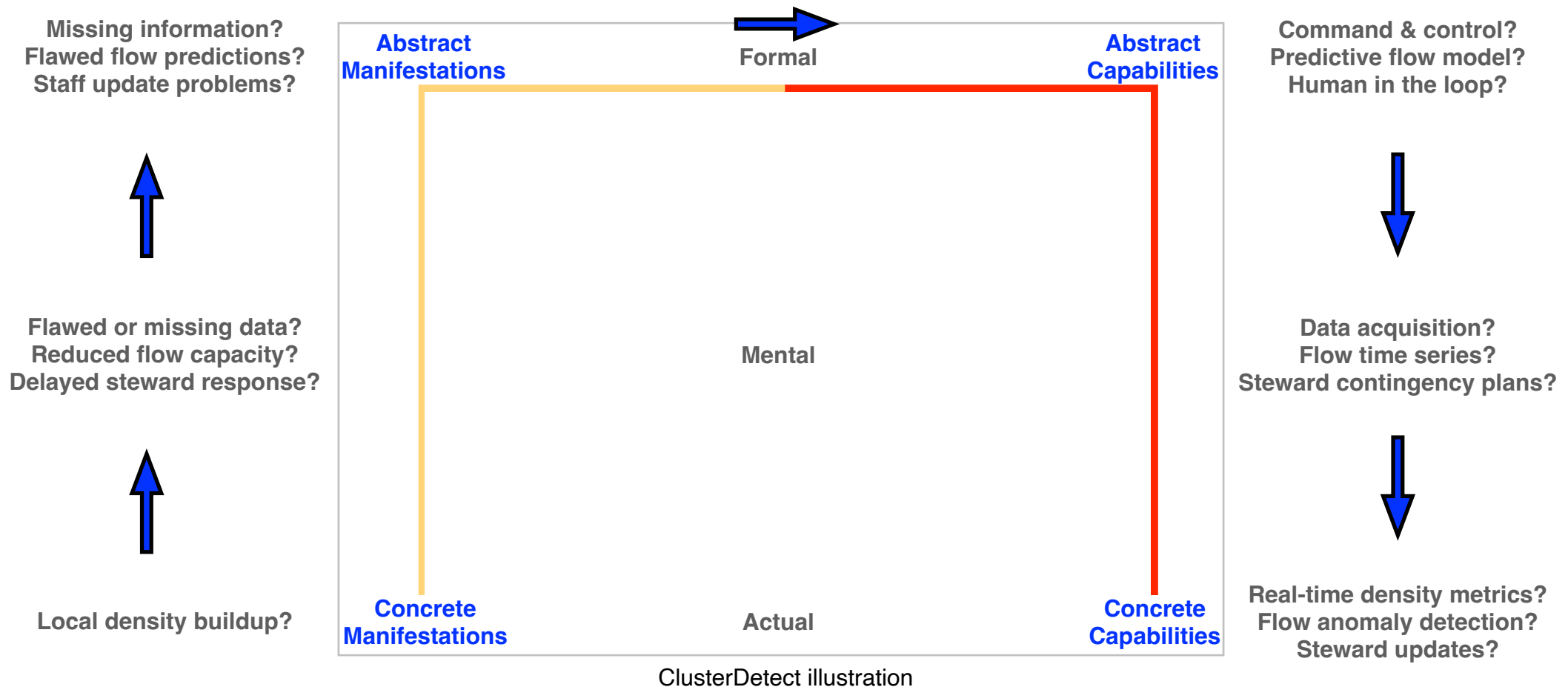
Manifestations



Manifestations are objects, actions, or events that reflect or embody the problem.

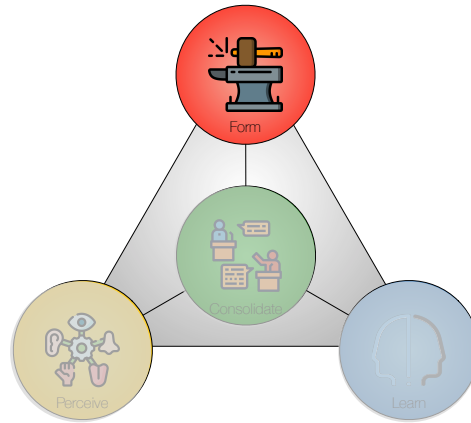
Concrete vs. Abstract

– Demand Pull

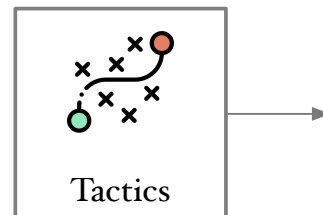
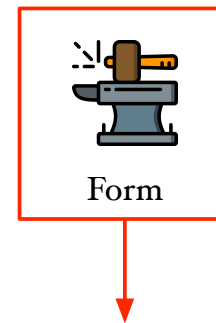


Pull—from problem toward solution. Making decisions on the way.

A problematic situation is noticed first. Designers identify manifestations, formulate the problem, and then develop a solution.



Capabilities

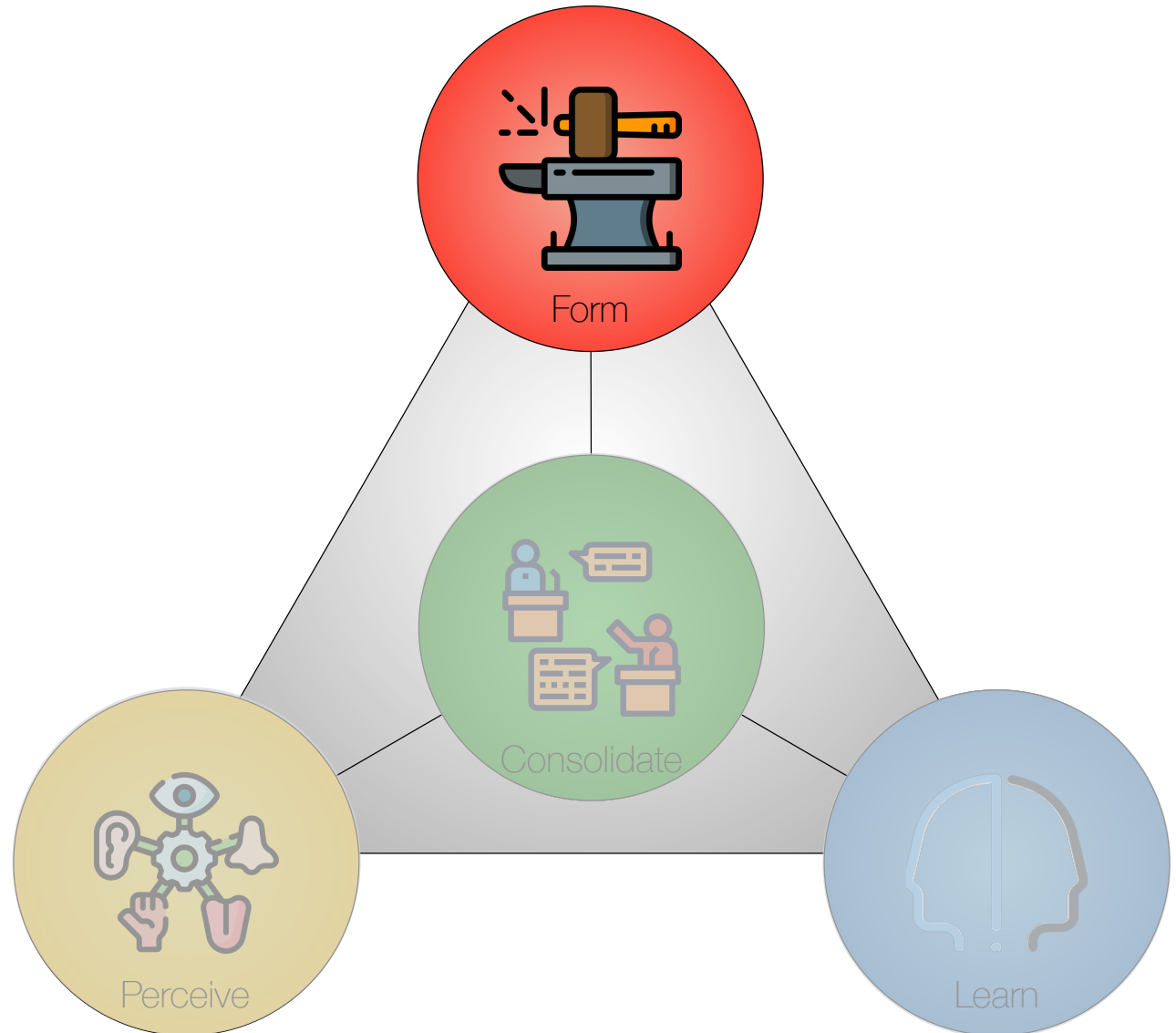


We define organic order as the kind of order that is achieved when there is a perfect balance between the needs of the parts, and the needs of the whole.

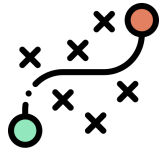
Christopher Alexander (1936–2022)

Four Core Activities

- *Perceive* – understanding the problem and its context.
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Form Terminology

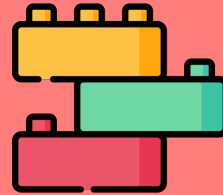


Capabilities



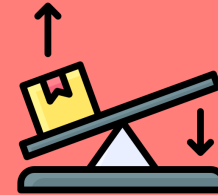
Capabilities are the features that are designed to address manifestations.

Inner Environment



The inner environment is what provides the capabilities.

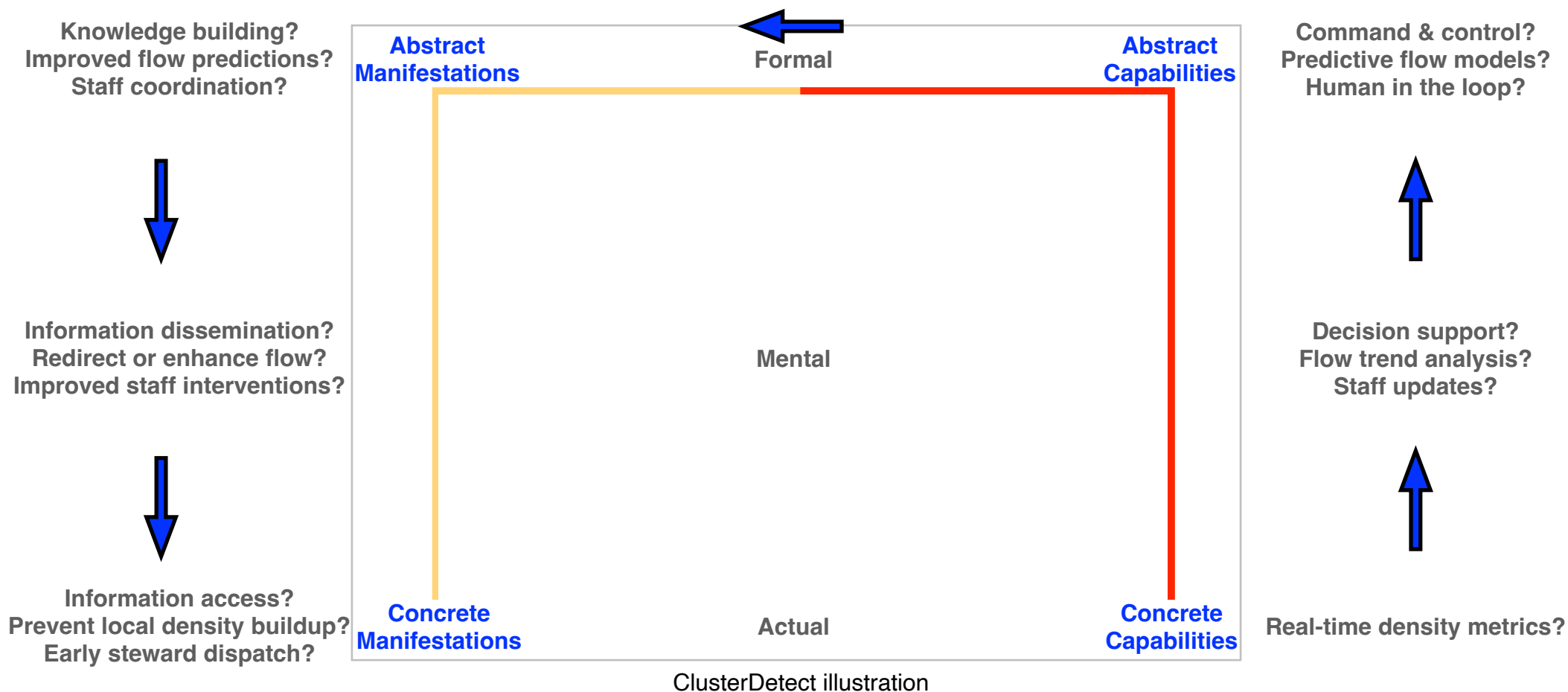
Leverage



Leverage points allow for the most valuable capabilities.

Concrete vs. Abstract

– Technology Push



Push—from solution toward problem

A technological possibility appears first. Designers explore what kinds of problems it may address, under what conditions, and for whom. What problematic situations could this help transform?

Problem-solving by Abstraction

- **Patterns** (Alexander, Gamma et al., Rising).
- **Metaphor use** (Dalsgaard, Beck).
- **Innovation Games** (Hohmann).
- **Lateral thinking** (De Bono).
- **SCAMPER** (Eberle).
 - Substitute, combine, adjust, modify, put to other uses, eliminate, reverse.
- **TRIZ** (Altshuller)
- and many more.

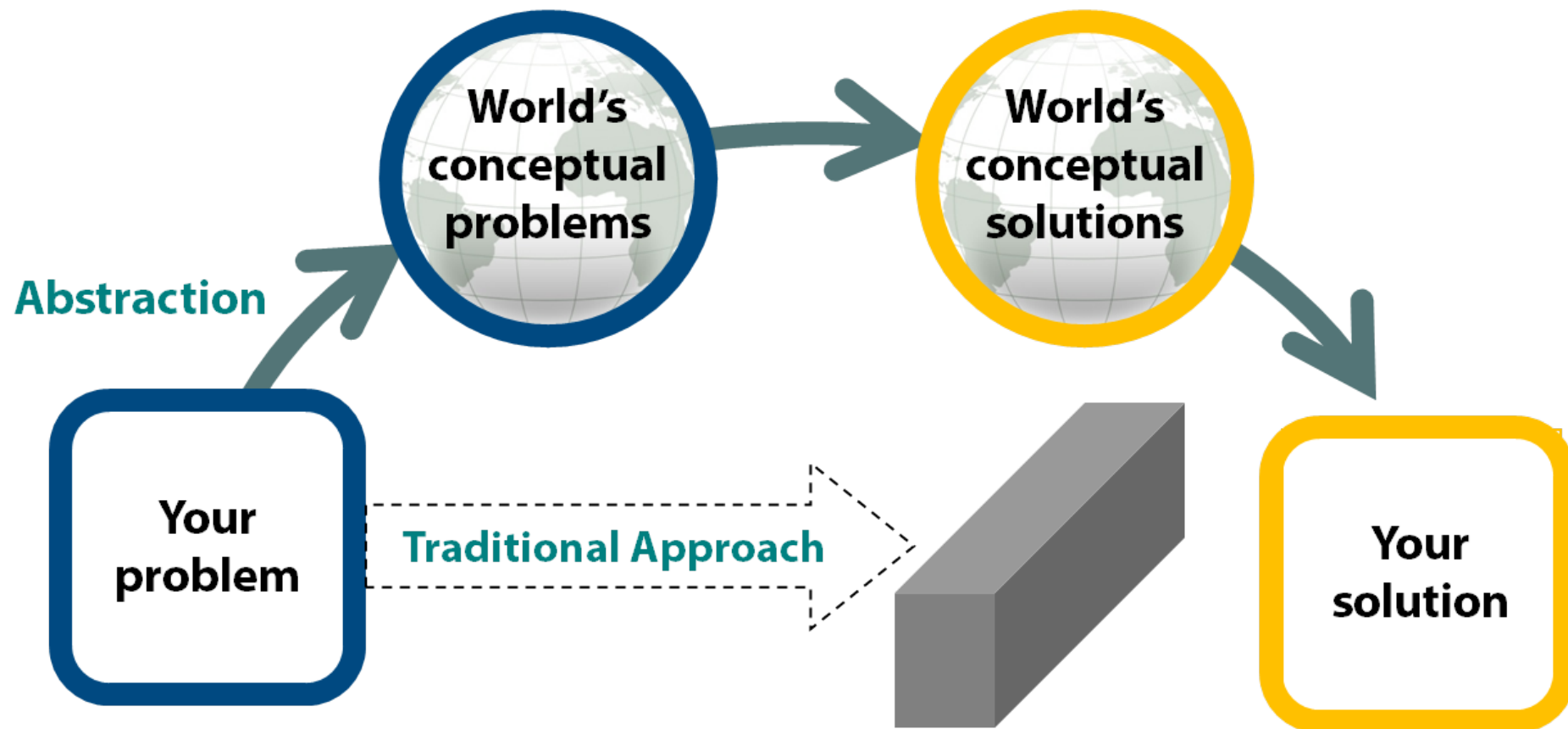
Patterns

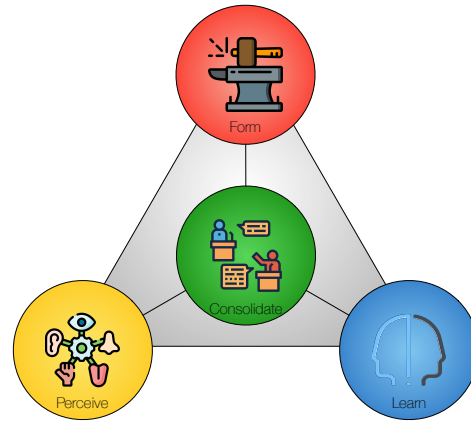
- [Alexander](#) (Architecture and general design).
- [Gamma et al](#) (Software design patterns).
- [Rising](#) (Application development).

40 TRIZ Principles

- | | | | |
|------------------------------|----------------------------------|-------------------------------|---------------------------------|
| 1 Segmentation | 2 Taking out | 3 Local quality | 4 Asymmetry |
| 5 Merging | 6 Universality | 7 Nested doll | 8 Anti-weight |
| 9 Preliminary anti-action | 10 Preliminary action | 11 Beforehand cushioning | 12 Equipotentiality |
| 13 The other way round | 14 Spheroidality - Curvature | 15 Dynamics | 16 Partial or excessive acti... |
| 17 Another dimension | 18 Mechanical vibration | 19 Periodic action | 20 Continuity of useful acti... |
| 21 Skipping | 22 *Blessing in disguise* o... | 23 Feedback | 24 Intermediary |
| 25 Self-service | 26 Copying | 27 Cheap short-living obje... | 28 Mechanics substitution |
| 29 Pneumatics and hydraul... | 30 Flexible shells and thin f... | 31 Porous materials | 32 Color changes |
| 33 Homogeneity | 34 Discarding and recoveri... | 35 Parameter changes | 36 Phase transitions |
| 37 Thermal expansion | 38 Strong oxidants | 39 Inert atmosphere | 40 Composite materials |

TRIZ Flowchart





Next Time

Next Exercise and Lecture

- Exercise 10:
 - Tactics, Manifestations & Capabilities.
- Lecture 11:
 - Chapter 18 in Essence: Merit

Exercise 10 (Tactics, Manifestations & Capabilities)

This exercise concerns the **tactics**, **manifestations**, and **capabilities** of your project.

- Will you characterize your approach as **push** or **pull**?
- Describe concrete and abstract **manifestations**.
- Describe concrete and abstract **capabilities**.
- Will your abstractions improve **evolvability** and make your design **future-proof**? How?